

Unit 1

Operating system services and components

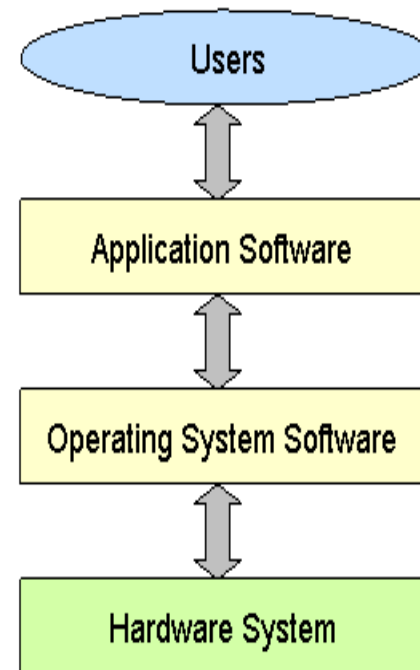


What is Operating System?

Definition:

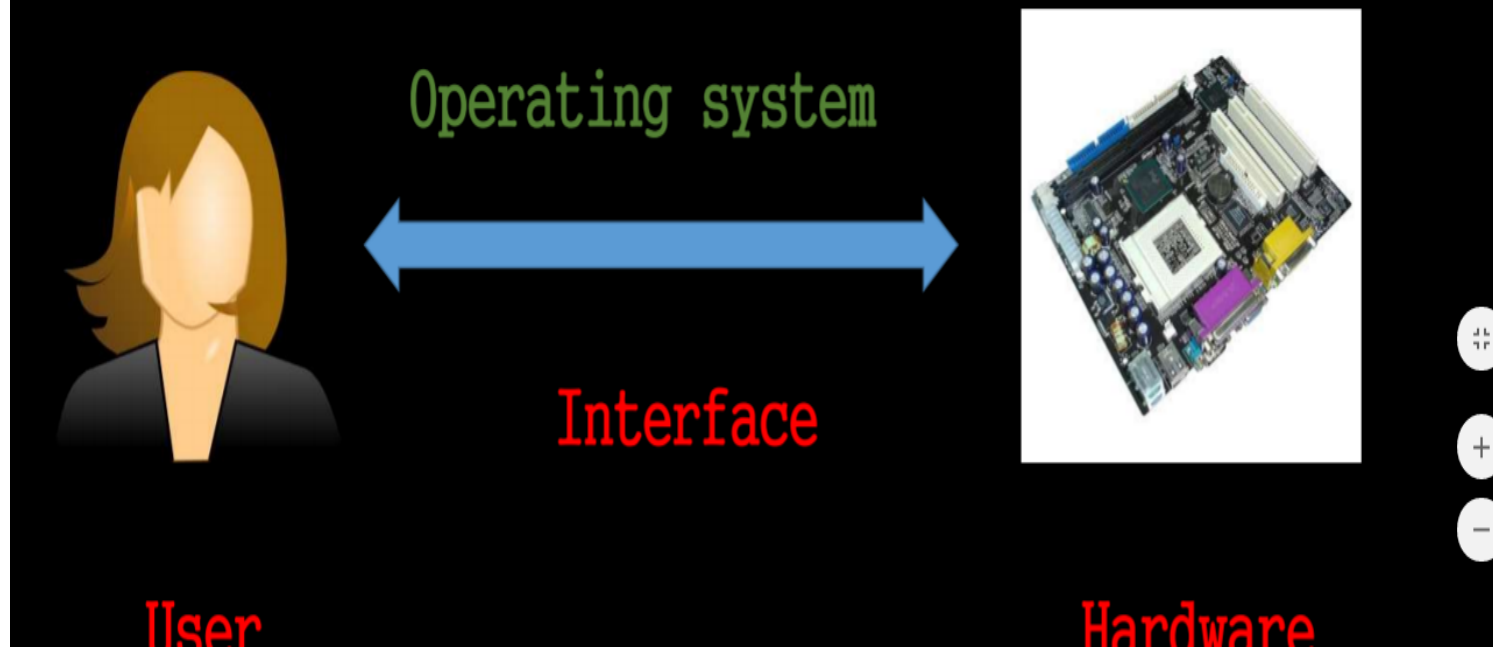
- An **operating system (OS)** is a collection of software that manages computer hardware resources .
- Without a computer operating system, a computer would be useless

“The operating system acts as an Interface Between the user and computer hardware”



What is Operating System?

Operating system is the interface between user and computer hardware



What is Operating System?

Main objective is to improve the performance and efficient use of computer resources.

Need of OS

We need an environment in which user can execute programs

Operating system goals:

- Execute user programs and make solving user problems easier
- Make the computer system convenient to use
- Use the computer resources in an efficient manner



Types of Softwares

- **System Softwares**

Program essential for the working of system. It is designed to run computer hardware and application softwares.

e.g. **Operating System**

Compiler, Assembler, Linker etc

- **Application software --**

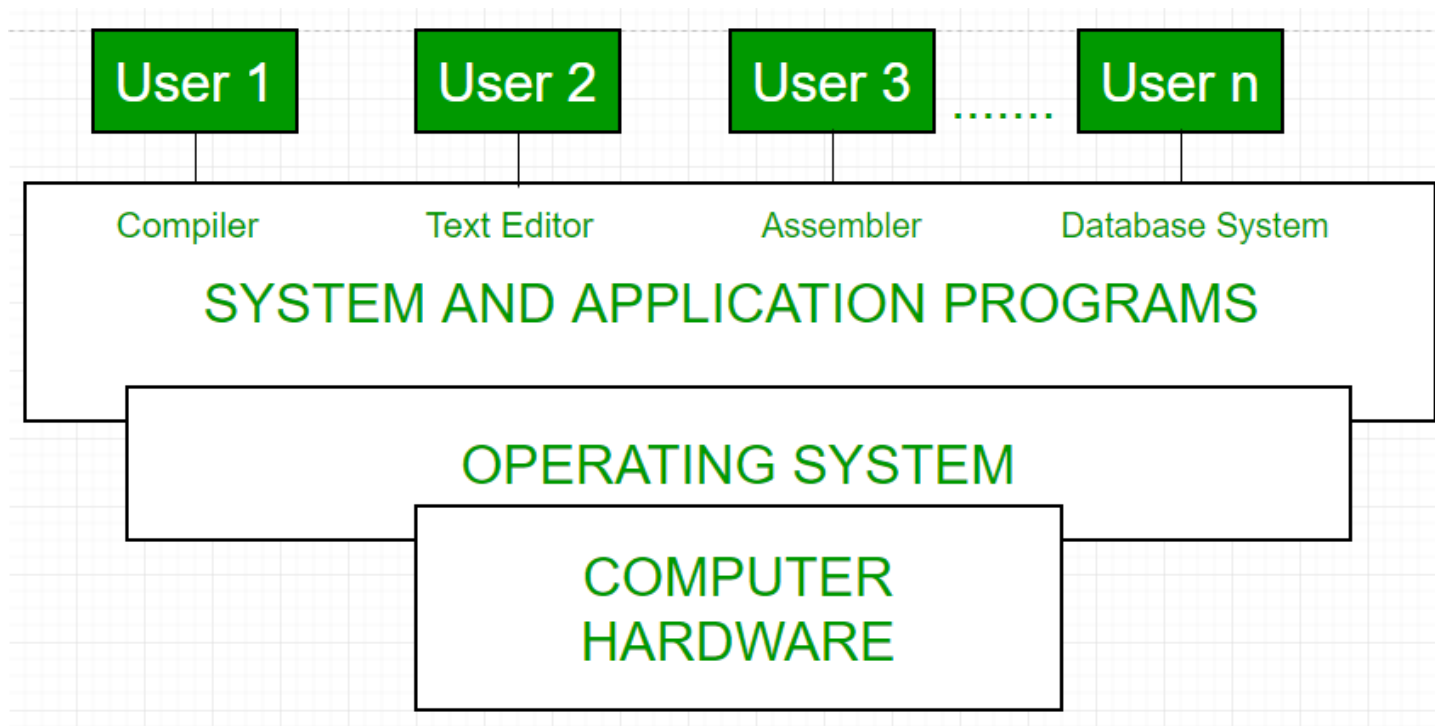
Group of programs designed for end users to perform specific task for the user.

e.g. word processor,

spreadsheet, accounting

application, web browser, media player, photo editor etc.





Conceptual view of a computer system



Assembler

- The input to an assembler is an assembly language program.
- The output is an object program plus information that enables the loader to prepare the object program for execution.
- By writing a series of ones and Zeros (Machine language), place them into the memory of the machine.



Compiler

- The High-level languages- examples are FORTRAN, COBOL, ALGOL and PL/I are processed by compilers and interpreters.
- A compiler is a program that accepts a source program in a “high-level language “and produces a corresponding object program.
- An interpreter is a program that appears to execute a source program



Loader

- A Loader is a routine that loads an object program and prepares it for execution.
- In general, the loader must load, relocate and link the object program.
- The loader is a program that places programs into memory and prepares them for execution.



Operations of OS

Start and shut down
a computer

Provide a user
Interface

Manage
Programs

Coordinate
Tasks

Configure
Devices

Manage
Memory

Establish An Internet
Connection

Control a
network

Provide
Utilities



Types of Interfaces

CLI (Command-Line Interface)

```

Loading CPM.SYS...

CP/M-86 for the IBM PC/XT/AT, Vers. 1.1 (Patched)
Copyright (C) 1983, Digital Research

Hardware Supported :

    Diskette Drive(s) : 3
    Hard Disk Drive(s) : 1
    Parallel Printer(s) : 1
    Serial Port(s) : 1
    Memory (Kb) : 640

D>a:
A>dir
A: PIP      CMD : STAT      CMD : SUBMIT  CMD : ASM86   CMD
A: GENCMD  CMD : D86      CMD : TOD      CMD : ED      CMD
A: HELP    CMD : HELP     HLP : SYS     CMD : ASSIGN  CMD
A: FORMAT  CMD : CLDIR    CMD : WRTLD8  CMD : BOOTPCBS SYS
A: BOOTWIN SYS : CPM      86 : WINSTALL SUB : PD      CMD
A: WCPM    SYS : DISKUTIL CMD

A>_
      User 0      0:00:11      Jan. 1, 2000

```

GUI (Graphical User Interface)



Functions of O.S

1. Controlling and allocating memory
2. Prioritizing System request
3. Controlling input and Output devices
4. Facilitating computer networking and managing files.
5. Transferring input from input devices to memory
6. Transferring input from memory to CPU
7. Transferring processed data from CPU to output devices
8. Show messages on screen in case of input and output devices



9. Display appropriate error messages by giving current status of peripherals and processes
10. Execution of programs and user instructions
11. Protection of storage devices from overwriting
12. Maintain files and directories with details
13. Provide security .



Different Types of Operating Systems

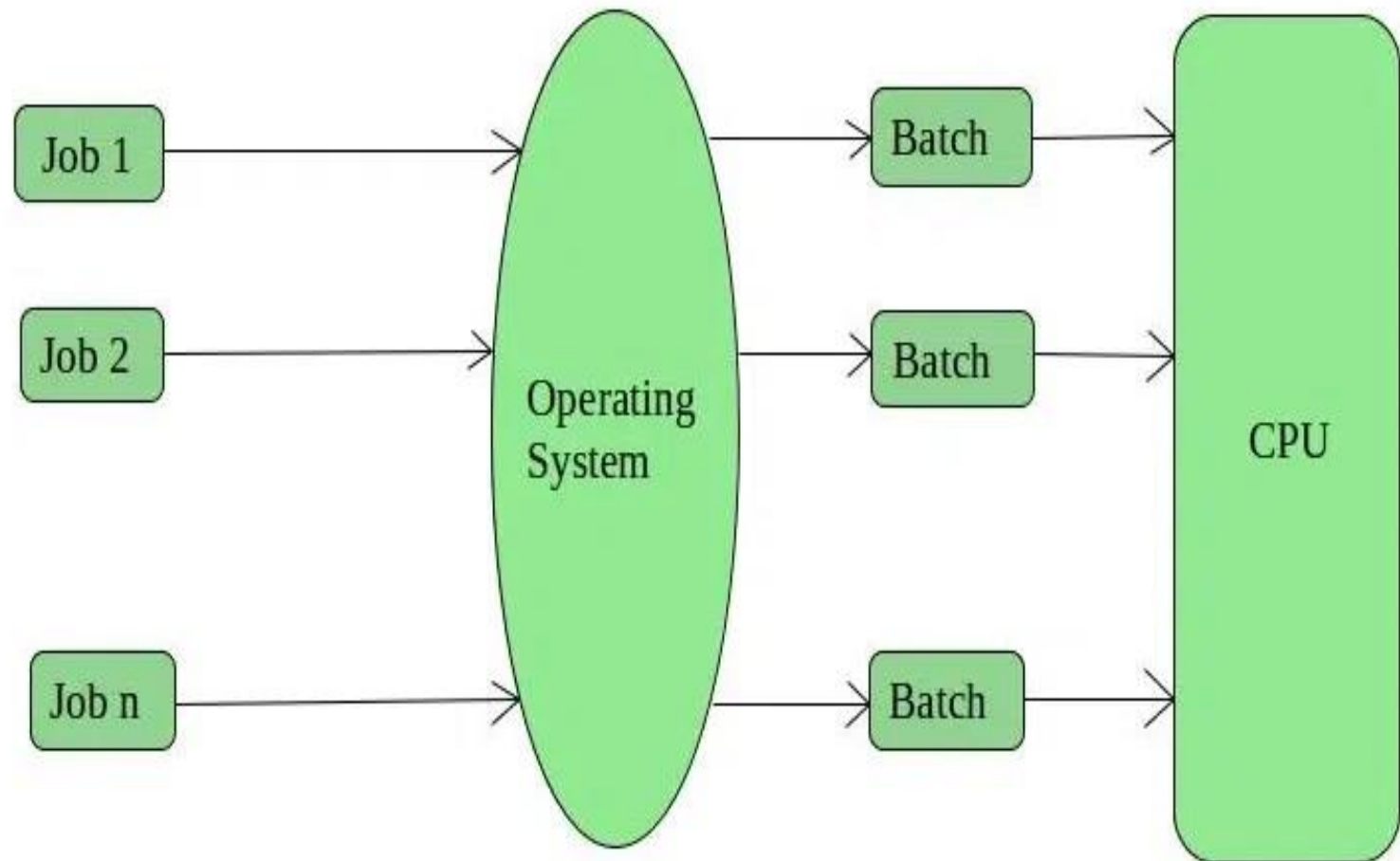
1. Batch Operating System
2. Multi-Programming Operating System
3. Multi-Processing Operating System
4. Distributed Operating System
5. Real-Time Operating System
6. Mobile Operating Systems



Batch Operating System

1. This type of operating system does not interact with the computer directly.
2. There is an operator which takes similar jobs having the same requirements and groups them into batches.
3. It is the responsibility of the operator to sort jobs with similar needs.
4. Batch Operating System is designed to manage and execute a large number of jobs efficiently by processing them in groups.





Advantages of Batch Operating System

1. Multiple users can share the batch systems.
2. The idle time for the batch system is very little.
3. It is easy to manage large work repeatedly in batch systems.

Disadvantages of Batch Operating System

1. CPU is not used efficiently. When the current process is doing IO, the CPU is free and could be utilized by other processes waiting.
2. The other jobs will have to wait for an unknown time if any job fails.
3. In a batch operating system, average response time increases as all processes are processed one by one.

Examples of Batch Operating Systems: Payroll Systems, Bank Statements, etc.

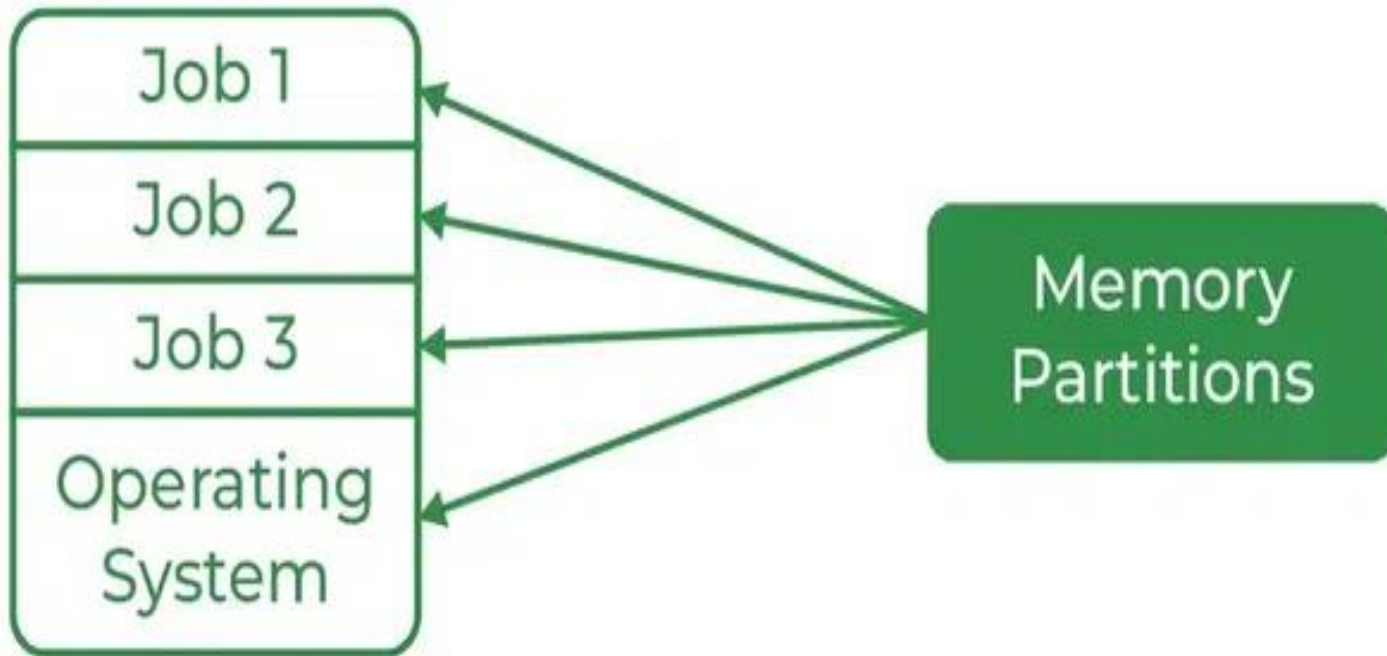


Multi-Programming Operating System

- Multiprogramming Operating Systems can be simply illustrated as more than one program is present in the main memory and any one of them can be kept in execution.
- This is used for better utilization of resources.



Multiprogramming



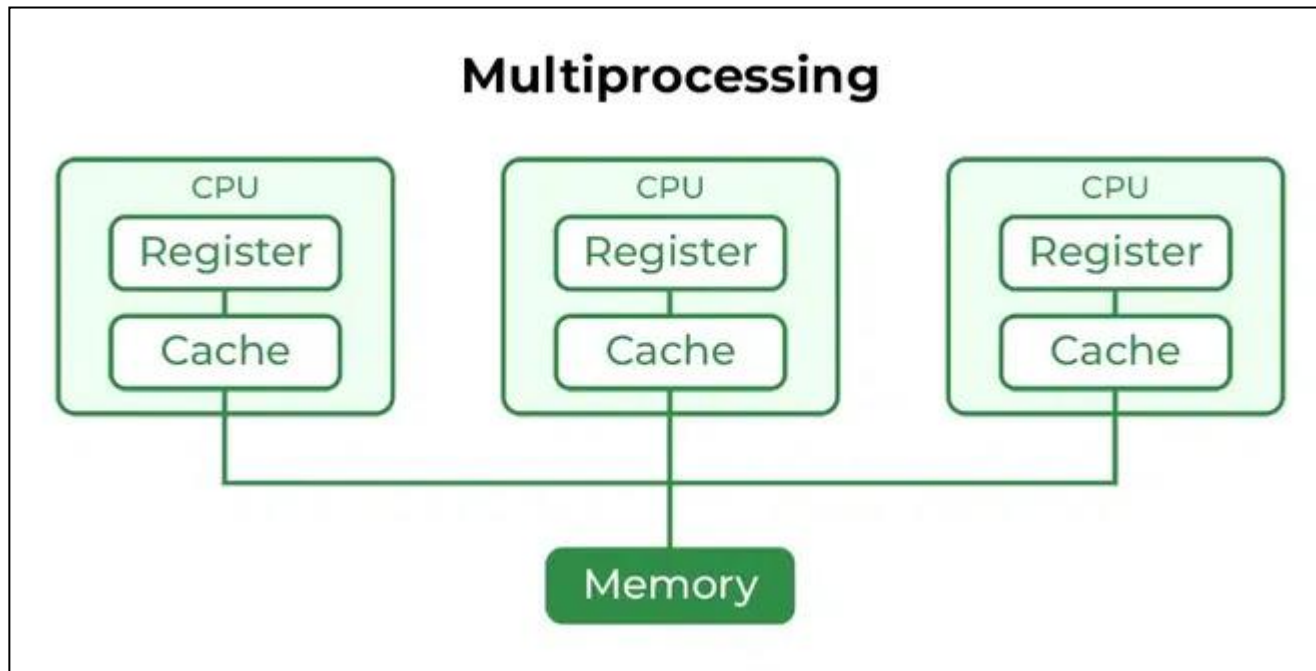
Advantages

1. CPU is better utilized, and the overall performance of the system improves.
2. It helps in reducing the response time.



Multi-Processing Operating System

A Multi-Processing Operating System is a type of Operating System in which more than one CPU is used for the execution of resources. It better the throughput of the System.

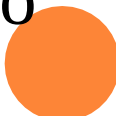


Advantages of a Multi-User Operating System

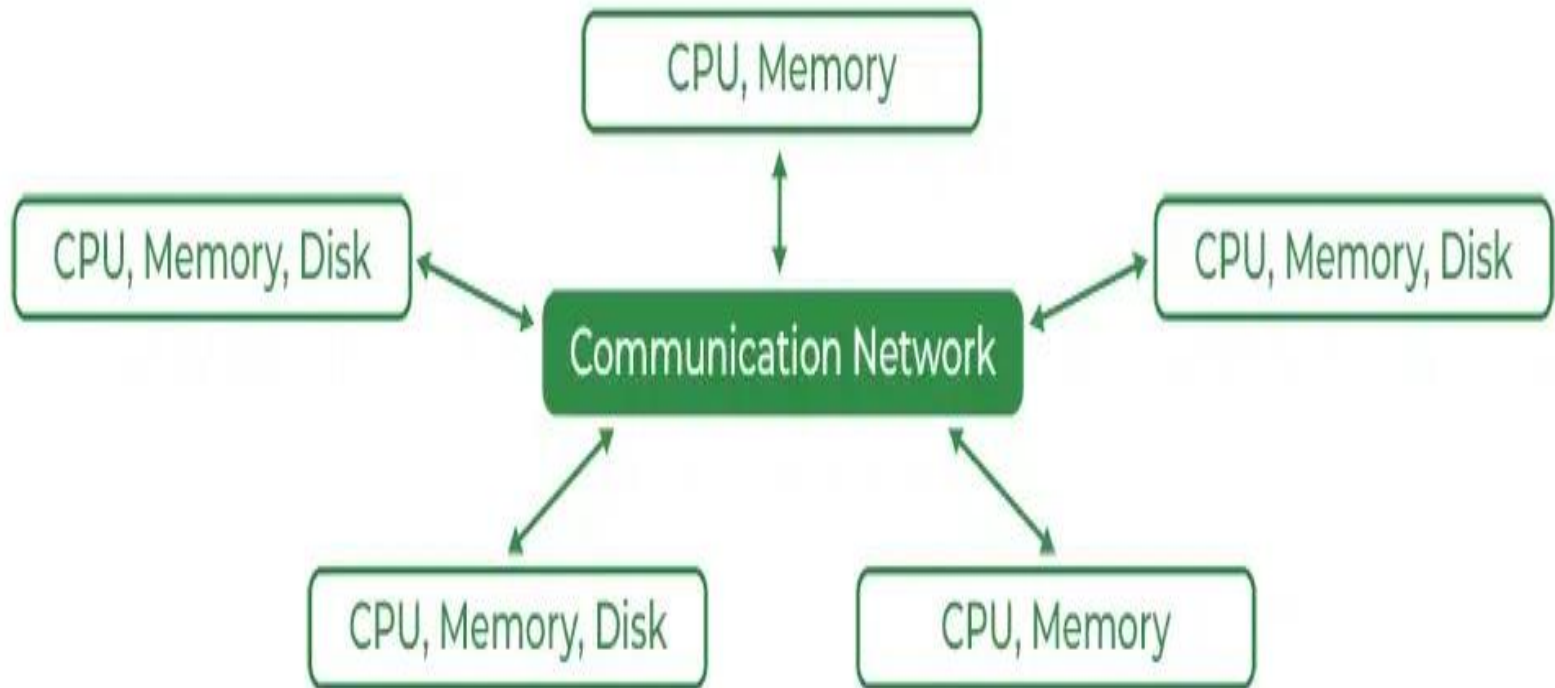
1. It increases the throughput of the system as processes can be parallelized.
2. As it has several processors, so, if one processor fails, we can proceed with another processor.



Distributed Operating System

1. Distributed Operating System is a type of model where applications are running on multiple computers linked by communications.
 2. Distributed OS runs on multiple **CPUs** but for an end-user, it is just an ordinary centralized operating system.
 3. It can share all resources like CPU, **disk**, **network interface**, nodes, **computers**, etc. from one site to another site, and it increases the data available on the entire system.
 4. All processors are connected by valid communication media such as high-speed buses and telephone lines, and in which every processor contains its own local memory along with other local processors.
 5. This operating system involves multiple computers, nodes, and sites, and these components are linked to each other with **LAN/WAN** lines.
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Architecture of Distributed OS



Advantages of Distributed Operating System

1. Failure of one will not affect the other network communication, as all systems are independent of each other.
2. Electronic mail increases the data exchange speed.
3. Since resources are being shared, computation is highly fast and durable.
4. Load on host computer reduces.
5. These systems are easily scalable as many systems can be easily added to the network.
6. Delay in data processing reduces.

Disadvantages of Distributed Operating System

1. Failure of the main network will stop the entire communication.
2. To establish distributed systems, the language is not yet well-defined.
3. These types of systems are not readily available as they are very expensive. Not only that the underlying software is highly complex and not understood well yet.



Real-Time Operating System

These types of OSs serve real-time systems.

The time interval required to process and respond to inputs is very small.

This time interval is called **response time**.

Real-time systems are used when there are time requirements that are very strict like missile systems, air traffic control systems, robots, etc.



Types of Real-Time Operating Systems

Hard Real-Time Systems: Hard Real-Time OSs are meant for applications where time constraints are very strict, and even the shortest possible delay is not acceptable.

These systems are built for saving lives like automatic parachutes or airbags which are required to be readily available in case of an accident.

Soft Real-Time Systems: These OSs are for applications where time is less strict.



Advantages of RTOS

Maximum Consumption: Maximum utilization of devices and systems, thus more output from all the resources.

Task Shifting: The time assigned for shifting tasks in these systems is very less. For example, in older systems, it takes about 10 microseconds to shift from one task to another, and in the latest systems, it takes 3 microseconds.

Focus on Application: Focus on running applications and give less importance to applications that are in the queue.

Real-time operating system in the embedded system: Since the size of programs is small, RTOS can also be used in embedded systems like in transport and others.

Error-Free: These types of systems are error-free.

Memory Allocation: Memory allocation is best managed in these types of systems.



Disadvantages of RTOS

Limited Tasks: Very few tasks run at the same time and their concentration is very less on a few applications to avoid errors.

Use heavy system resources: Sometimes the system resources are not so good and they are expensive as well.

Complex Algorithms: The algorithms are very complex and difficult for the designer to write.

Examples of Real-Time Operating Systems are Scientific experiments, medical imaging systems, industrial control systems, weapon systems, robots, air traffic control systems, etc.



Mobile Operating Systems

Mobile operating systems are designed specifically for mobile devices such as smartphones and tablets.

Examples of such operating systems are Android and iOS.

These operating systems manage the hardware and software resources of the device, providing a platform for running applications and ensuring a seamless user experience.



Advantages of Mobile Operating Systems

User-Friendly Interfaces: Mobile operating systems are designed to be intuitive and easy to use, making them accessible to a wide range of users.

Extensive App Ecosystems: The availability of a vast number of applications allows users to customize their devices to meet their specific needs.

Connectivity Options: Mobile operating systems support multiple connectivity options, enabling users to stay connected wherever they go.

Regular Updates: Mobile operating systems receive regular updates, including new features, security patches, and performance improvements.



Disadvantages Mobile Operating Systems

Battery Life Constraints: Despite advancements in power management, battery life remains a challenge for mobile devices, especially with heavy usage.

Security Risks: Mobile devices are susceptible to various security threats, such as malware and phishing attacks, which can compromise user data.

Fragmentation: In the case of Android, the wide range of devices and customizations can lead to fragmentation, making it difficult for developers to ensure compatibility across all devices.

Limited Hardware Resources: Mobile devices have limited processing power, memory, and storage compared to desktop computers, which can affect the performance of resource-intensive applications.

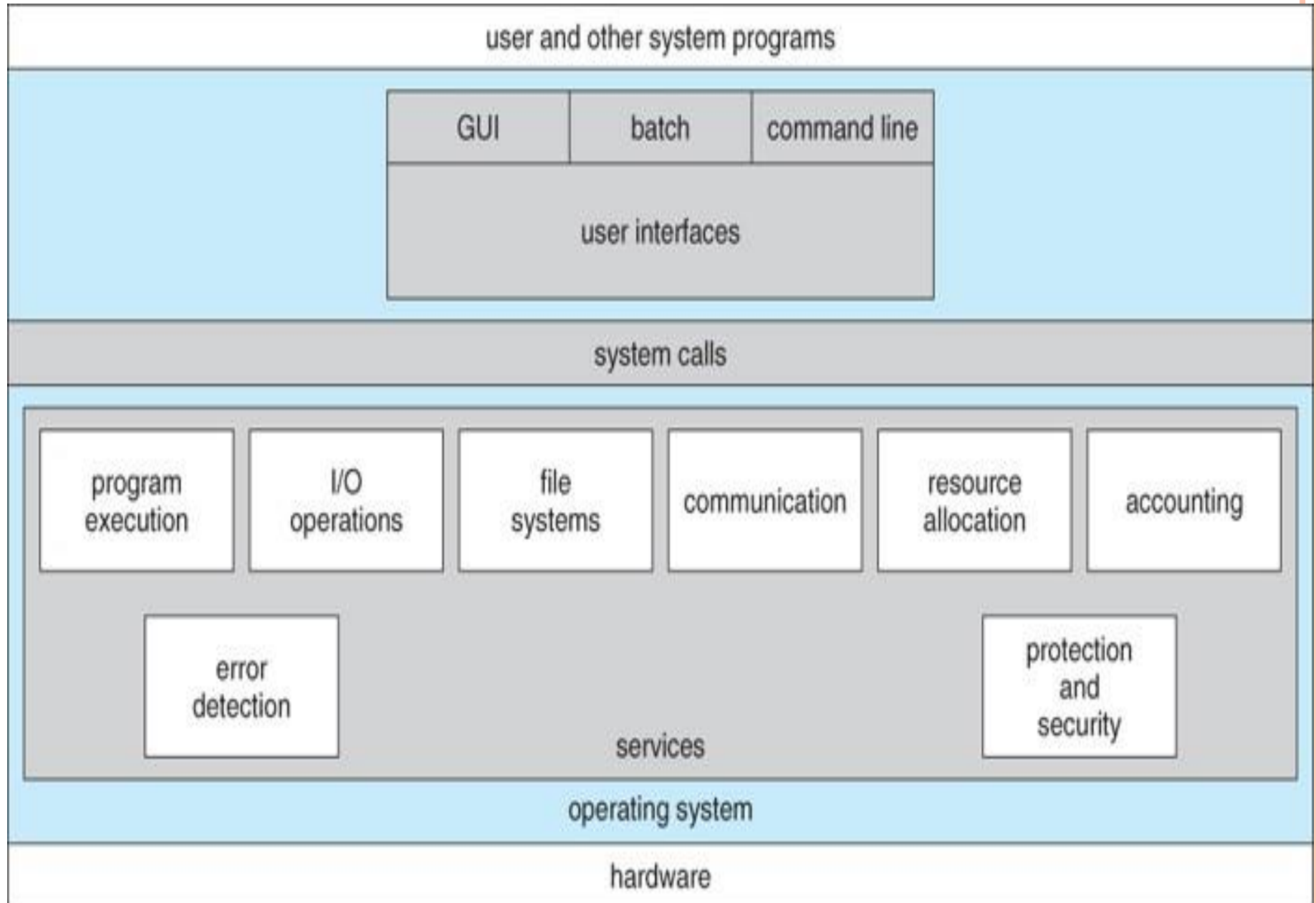


SERVICES OF OPERATING SYSTEM

- An Operating System supplies different kinds of services to both the users and to the programs as well.
- Following are the common services offered by an operating systems:
 1. User Interface
 2. Program Execution
 3. File system manipulation
 4. Input / Output Operations
 5. Communication
 6. Resource Allocation
 7. Error Detection
 8. Accounting
 9. Security and protection



DIAGRAM OF OS SERVICES



1. USER INTERFACE

- User interface is a medium through which user interacts with the system.
- It is categorized in three types:
 - **Command line interface (CLI)** usually deals with using text commands and a technique for entering those commands.
 - **Batch interface (BI):** commands and directives are used to manage those commands that are entered into files and those files get executed.
 - **Graphical user interface (GUI):** which is a window system with a pointing device (like mouse or trackball) to point to the I/O, choose from menus driven interface and to make choices viewing from a number of lists and a keyboard to entry the texts.



PROGRAM EXECUTION

- The operating system must have the capability to load a program into memory and execute that program.
- Furthermore, the program must be able to end its execution, either normally or abnormally / forcefully.



FILE SYSTEM MANIPULATION

- Programs need has to be read and then write them as files and directories.
- File handling portion of operating system also allows users to create and delete files by specific name along with extension, search for a given file and / or list file information.



I/O OPERATIONS

- A program which is currently executing may require I/O, which may involve file or other I/O device.
- For efficiency and protection, users cannot directly govern the I/O devices.
- So, the OS provide a means to do Input / Output operation which means read or write operation with any file.



COMMUNICATION

- Process needs to swap over information with other process.
- Processes executing on same computer system or on different computer systems can communicate using operating system support.
- Communication between two processes can be done using shared memory or via message passing.



RESOURCE ALLOCATION

- When multiple jobs running concurrently, resources must need to be allocated to each of them.
- Resources can be CPU cycles, main memory storage, file storage and I/O devices.
- CPU scheduling routines are used here to establish how best the CPU can be used.



ERROR DETECTION

- Errors may occur within CPU, memory hardware, I/O devices and in the user program.
- For each type of error, the OS takes adequate action for ensuring correct and consistent computing.



ACCOUNTING

- This service of the operating system keeps track of which users are using how much and what kinds of computer resources have been used for accounting or simply to accumulate usage statistics.



PROTECTION AND SECURITY

- Protection includes in ensuring all access to system resources in a controlled manner.
- For making a system secure, the user needs to authenticate him or her to the system before using (usually via login ID and password).



SYSTEM CALL

- The interface between a process and an operating system is provided by system calls.
- System calls are usually made when a process in user mode requires access to a resource. Then it requests the kernel to provide the resource via a system call.

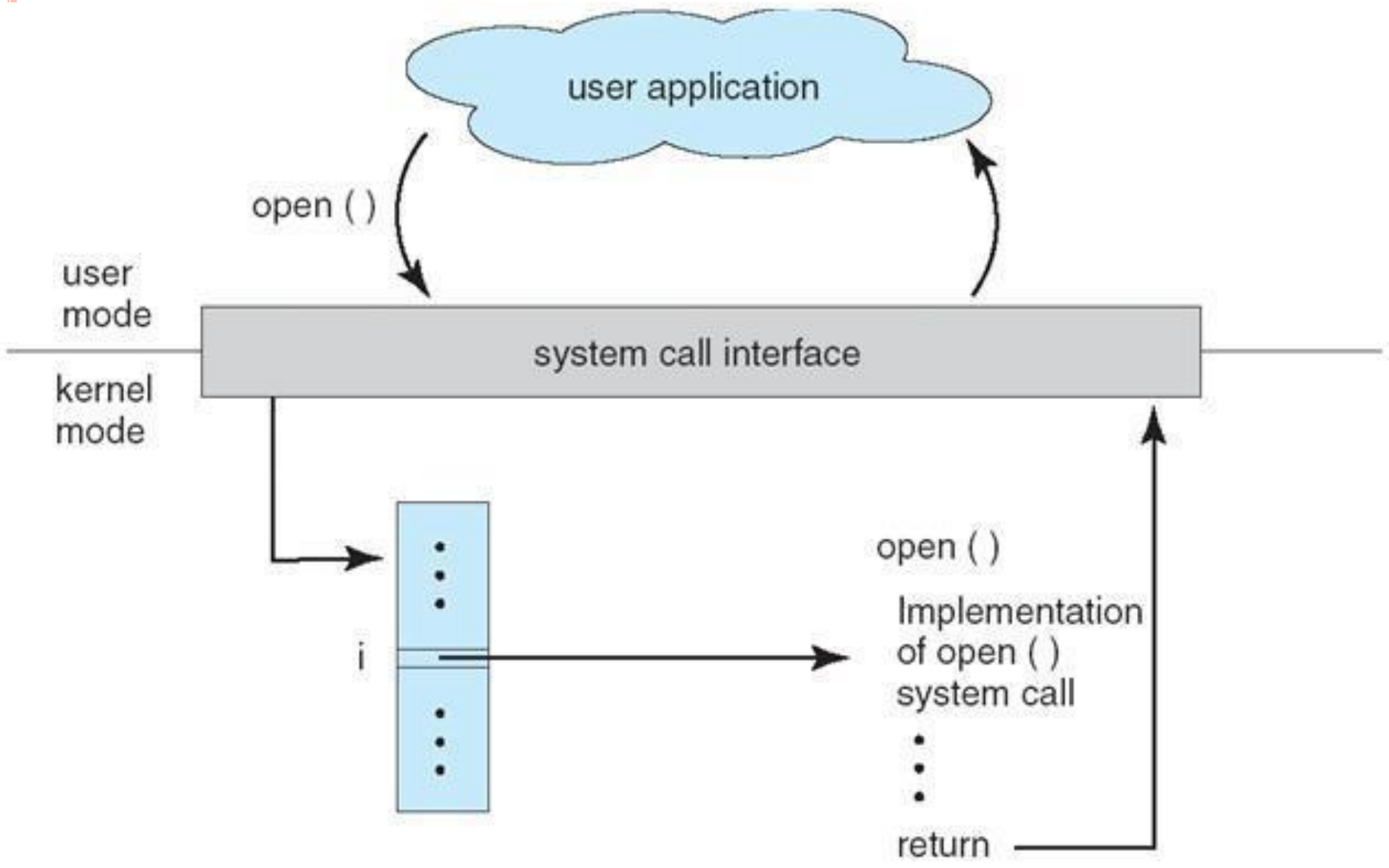


SYSTEM CALL

- Typically written in a high-level language (C or C++)
- Mostly accessed by programs via a high-level Application Program Interface (API)
- Three most common APIs are
 - Win32 API for Windows,
 - POSIX API (all versions of UNIX, Linux, and Mac OS X), and
 - Java API for the Java virtual machine (JVM)



SYSTEM CALL DIAGRAM(IMPLEMENTATION OF OPEN() SYSTEM CALL



SYSTEM CALL

- User creates application program with the help of system call interface.
- System call interface converts user instructions into appropriate system call.
- It also maintains a dispatch table consisting system call and its index number.
- After executing user instruction system call interface provides appropriate system call and indexed number.
- Then trap occurs the os kernel. kernel determines system call according to index number.
- Execution of the system call implements in kernel mode and reply is forwarded to the system call interface and system call interface to the application program of the user.



SYSTEM CALL

- There are 5 different categories of system calls –
 - **Process control**
 - **File management**
 - **Device management**
 - **Information maintenance**
 - **Communication**



PROCESS CONTROL

- end, abort
- load, execute
- create process, terminate process
- get process attributes, set process attributes
- wait for time
- wait event, signal event
- allocate and free memory



FILE MANAGEMENT

- create file, delete file
- open, close
- read, write, reposition
- get file attributes, set file attributes



Device management

- request device, release device
- read, write, reposition
- get device attributes, set device attributes
- logically attach or detach devices



INFORMATION MAINTENANCE

- get time or date, set time or date
- get system data, set system data
- get process, file, or device attributes
- set process, file, or device attributes



COMMUNICATION

- create, delete communication connection
- send, receive messages
- transfer status information
- attach or detach remote devices



OPERATING SYSTEM COMPONENTS

- Following are the components of OS:

- Process management
- I/O management
- Main Memory management
- File Management
- Secondary Storage Management
- Program Management
- Resource Management
- Security And Protection



PROCESS MANAGEMENT

- A process is a program in execution.
- A process needs certain resources, including CPU time, memory, files, and I/O devices, to accomplish its task. „
- The operating system is responsible for the following activities in connection with process management.
 - Process creation and deletion.
 - process suspension and resumption.
 - provision of mechanisms for: •process synchronization •process communication



I/O MANAGEMENT

- Provide an abstract level of H/W devices and keep the details from applications to ensure proper use of devices, to prevent errors, and to provide users with convenient and efficient programming environment.
- The I/O system consists of:
 - A buffer-caching system
 - A general device-driver interface
 - Drivers for specific hardware device
- Activities:
 - Hide the details of H/W devices
 - Manage main memory for the devices using cache, buffer, and spooling
 - Maintain and provide device driver interfaces



MAIN MEMORY MANAGEMENT

- Memory is a large array of words or bytes, each with its own address.
- It is a repository of quickly accessible data shared by the CPU and I/O devices.
- Main memory is a volatile storage device. It loses its contents in the case of system failure.
- The operating system is responsible for the following activities in connections with memory management:
 - Keep track of which parts of memory are currently being used and by whom.
 - Decide which processes to load when memory space becomes available.
 - Allocate and deallocate memory space as needed



FILE MANAGEMENT

- A file is a collection of related information defined by its creator.
- Commonly, files represent programs (both source and object forms) and data. „
- The operating system is responsible for the following activities in connections with file management:
 - File creation and deletion.
 - Directory creation and deletion.
 - Support of primitives for manipulating files and directories.
 - Mapping files onto secondary storage.
 - File backup on stable (nonvolatile) storage media.



SECONDARY STORAGE MANAGEMENT

- Since main memory (primary storage) is volatile and too small to accommodate all data and programs permanently, the computer system must provide secondary storage to back up main memory.
- Most modern computer systems use disks as the principle on-line storage medium, for both programs and data. „
- The operating system is responsible for the following activities in connection with disk management:
 - Free space management
 - Storage allocation
 - Disk scheduling



SECURITY AND PROTECTION

- Protection refers to a mechanism for controlling access by programs, processes, or users to both system and user resources. „
- Security ensures that avoid unauthorized access.
- The Activities of protection and security mechanism :
 - distinguish between authorized and unauthorized usage.
 - specify the controls to be imposed.
 - provide a means of enforcement.



PERFORMANCE MONITOR

○ top command :

- Linux top command is performance monitoring program which is used frequently by many system admin to monitor linux performance.
- top command is used to display all the running and active real-time processes in ordered list and update it regularly.
- It display cpu usage,memory usage,swap memory,cache,size,buffer size,process id,user.
- top command is much useful for system administrator to monitor and take correct action when required.
- Syntax:top



PERFORMANCE MONITOR

- vmstat:
 - Used to display statistics of virtual memory, kernel threads ,disk ,system processes, i/o blocks,interrupts,cpu activity and much more.
 - Syntax:vmstat

